

• First Fit → Selects the first suitable free block.

5:02 PM

4G 69

• Worst Fit → Selects the largest free block.

=



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Easy point wise

## First Fit Memory Allocation

- It is a memory allocation technique in which the process is placed in the **first free memory block** that is large enough.
- It is simple and fast.
- It may cause memory wastage due to fragmentation.

## Worst Fit Memory Allocation

- It is a memory allocation technique in which the process is placed in the **largest available memory block**.
- It leaves larger free spaces after allocation.
- It may reduce the chance of small unused spaces.



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Logical address and physical address

## Logical Address and Physical Address in Operating System

### Logical Address

- A **logical address** is the address generated by the CPU while a program is running.
- It is also called a **virtual address**.
- The user/program can see this address.
- It is converted into a physical address by the memory management unit (MMU).

### Physical Address

- A **physical address** is the actual address location in the main memory (RAM).
- It is the address used by the hardware to access data.
- It is not directly visible to the user.



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## Preemptive Scheduling

- In this scheduling method, the operating system can **interrupt a running process** and give the CPU to another process.
- A process can be stopped before completing its task.
- It improves response time.
- It is used in time-sharing systems.

**Examples:** Round Robin, Shortest Remaining Time First (SRTF), Priority Scheduling.

## Non-Preemptive Scheduling

- In this scheduling method, a process keeps the CPU until it **completes its execution** or enters a waiting state.
- The operating system cannot forcefully stop the running process.
- It is simple and has less overhead.

**Examples:** First Come First Serve (FCFS), Shortest Job First (SJF).



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## IPC (Inter Process Communication)

### Definition:

IPC is a method used in an operating system that allows **different processes to communicate and share information** with each other.

### Types of IPC:

#### 1. Shared Memory

- Processes share the same memory area.
- It allows fast communication.

#### 2. Message Passing

- Processes send and receive messages to communicate.

### Advantages:

- Helps processes work together.
- Allows data sharing between processes.
- Improves system performance.

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### Points:

- It helps prevent conflicts when multiple processes access the same resource.
- It allows only a limited number of processes to use a resource at a time.
- It is used to solve the critical section problem.

### Types of Semaphore:

#### 1. Binary Semaphore

- Has only two values: 0 and 1.
- Used for mutual exclusion.

#### 2. Counting Semaphore

- Can have multiple values.
- Used to control access to multiple resources.

### Example:

A semaphore can control access to a printer so that only one process uses it at a time.



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## Thrashing

- Thrashing is a condition where the operating system spends more time **swapping pages** than executing processes.
- It occurs due to insufficient memory and causes slow system performance.

## Virtual Memory

- Virtual memory is a memory management technique that uses **secondary storage as an extension of RAM**.
- It allows large programs to run even when the physical memory is limited.



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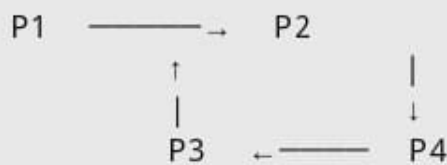


## Wait-for Graph

### Definition:

A **Wait-for Graph** is a graph used to detect **deadlock** in an operating system. It shows which process is waiting for another process to release a resource.

### Diagram Example:



### Explanation:

- P1 is waiting for P2
- P2 is waiting for P4
- P4 is waiting for P3
- P3 is waiting for P1

This creates a **cycle**, so a **deadlock occurs**.



### Points:

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Explain any 2 directory structure

A **directory structure** is the way files and folders are organized on a computer system. Two common types are:

## 1. Single-Level Directory Structure

- All files are stored in one single directory (folder).
- It is simple and easy to manage for small systems.
- File names must be unique because there is only one directory.

**Example:**



```
Root Directory
|
├── file1.txt
├── file2.txt
├── image.jpg
└── program.exe
```

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• Simple design





### Advantages:

- Simple design
- Easy to implement

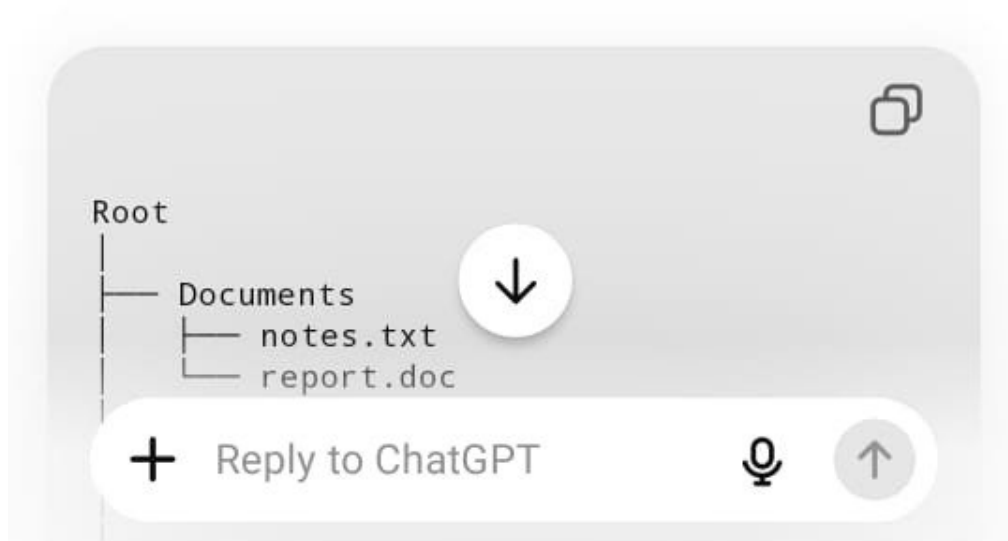
### Disadvantages:

- Difficult to manage many files
- Name conflicts can occur

## 2. Tree Directory Structure

- Files are organized in a hierarchy of directories and subdirectories.
- It is the most commonly used structure in modern operating systems.

### Example:





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Example:

```
graph TD; Root --> Documents; Root --> Pictures; Root --> Music; Documents --> notes_txt[notes.txt]; Documents --> report_doc[report.doc]; Pictures --> photo_jpg[photo.jpg]; Music --> song_mp3[song.mp3];
```

**Advantages:**

- Better organization of files
- Easier searching and management
- Allows users to group related files

**Disadvantages:**

- More complex than a single-level directory structure
- Requires more management



These are two basic directory structures used in operating systems.



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- **SCAN (Disk Scheduling):**

- Also called **Elevator algorithm**.
- Disk arm moves in one direction servicing requests, then reverses direction.

- **C-SCAN (Circular SCAN):**

- Disk arm moves in only **one direction** servicing requests.
- After reaching the end, it jumps back to the beginning without servicing.

- **LOOK:**

- Similar to SCAN.
- The disk arm goes only up to the **last pending request** in that direction, not the end of the disk.

- **C-LOOK:**

- Similar to C-SCAN.
- Moves in one direction up to the last request, then returns to the first request.

- **FIFO (First In First Out):**

- The first process,  that enters is removed first.



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• **SSTF / C-SSTF (Shortest Seek Time First):**



- **FIFO (First In First Out):**
  - The first process/page that enters is removed first.
  - Uses a simple queue method.
- **STF / SSTF (Shortest Seek Time First):**
  - Selects the disk request closest to the current head position.
  - Reduces seek time compared to FIFO.
- **LRU (Least Recently Used):**
  - Removes the page that has not been used for the longest time.
  - Used in page replacement algorithms.
- **Optimal Page Replacement:**
  - Replaces the page that will not be used for the longest time in the future.
  - Gives the minimum number of page faults.
- **SCAN (again):**
  - Services disk requests in a fixed direction and then changes direction like an elevator.



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